

Presentation on

The problems with the rapid **EV Charging** and the future challenges

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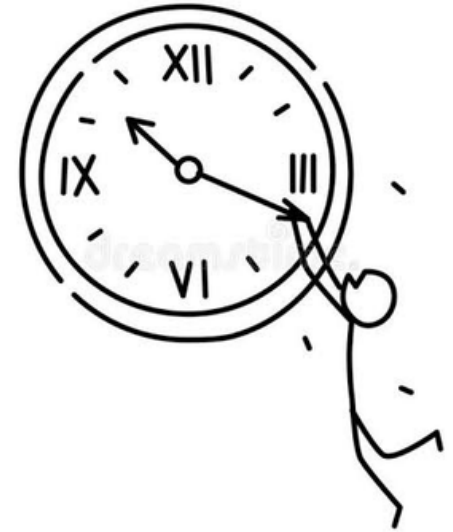
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Why we need the rapid EV charging

- Conventional EV charging can take several hours.
- Peoples do no have time and patience
- Make EVs More Practical for Long-Distance Travel
- Reduce Vehicle Downtime for Commercial Fleets



Does we have technology for rapid charging

- High-C-rate Battery Technology
- VOOC charging Developed by OPPO

Today, smartphones can already achieve around 0–80% charging in roughly 10–20 minutes using technologies such as OPPO SuperVOOC and Xiaomi HyperCharge



Charging times and limited ranges have been a major factor constraining the switch from gas and diesel vehicles to EVs, though Chinese drivers have [embraced that change](#), with sales of battery powered and hybrid vehicles jumping 40% last year.

Home > News International > StoreDot and Polestar showcase world's first 10-minute EV charge

StoreDot and Polestar showcase world's first 10-minute EV charge

Successful charging of a 77kWh battery pack powered by StoreDot's extreme fast charging high energy (300Wh/kg) silicon-dominant battery cells, installed in a fully driveable Polestar 5 prototype, from 10% to 80% in under 10 minutes.

By Autocar Professional Bureau 30 Apr 2024 4184 Views

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We have *need* and the
technology then what is
the problem



Problem -

Rapid charging need very high power in very low duration of time

Example

Battery capacity = 70 kWh
Charging = 0% → 80%
Charging time = 10 minutes

Energy required
 $E = 56 \text{ kWh}$

Convert time
 $10/60 = 0.1667 \text{ hr}$

Power = $56/0.1667$
Power = 336 kW

What happens with multiple cars?

$$10 \times 373 = 3730 \text{ kW}$$

Power required is approx 3.7 MW

3.7 MW is roughly equivalent to the simultaneous load of about 2,500 households at 1.5 kW each

Ten EVs charging simultaneously can demand around 3.7 MW—equivalent to the instantaneous rated output of roughly 7,400 solar panels rated at 500 W each.

A 70 kWh EV would require an average of approximately 336 kW to charge from 0% to 80% in 10 minutes, assuming ideal 100% efficiency.

- Grid can not handle that much of sudden variable load
- Need very big transformers to deliver that much amount of power

For 3.73 MW load -

Current = $3.73 \times 10^6 / 1000$

Current = 3730A

Current one car needed = 373 A

Load	Current @ 1000 V DC	Approx. copper area
1 EV	373 A	~75 mm ²
10 EVs	3,730 A	~746 mm ²



~75 mm²

For a typical 132 kV overhead transmission line in India, conductors are often in the range of roughly 150–300 mm² aluminum-based conductor cross-sectional area

A 3.73 MW EV charging station at 1000 V can require about 3,730 A. A simple 5 A/mm² estimate gives ~746 mm² of copper cross-sectional area—equivalent to a solid copper conductor roughly **31 mm in diameter**.

Generally the large transformer is made to deliver power at very high voltage level like 33kv but here we need to deliver current at very low voltage like 800V it means transformer it self has to deliver very high current it mean the transformer winding cross-section area shoul very big approx 3 cm in diameter

Ultra-fast charging doesn't just demand more electrical power—it also converts part of that power into heat in transformers, cables, busbars and connectors. The higher the current, the more difficult and important thermal management becomes.



THANK YOU !